

Regular/Supplementary Winter Examination – 2024

Course: B.TECH

Branch : Electrical Engineering and Allied

Semester : V

Subject Code & Name: BTEEC501 Power System Analysis

Max Marks: 60

Date:06/02/2025

Duration: 3 Hr.

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per CBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculator is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The load flow solution is always assured in case of:	CO1	1
	(a) Newton-Raphson method (b) Gauss method (c) Gauss-Seidel method (d) None of these		
	The p.u. impedance value of an alternator corresponding to base values 13.2 kV and 30 MVA is 0.2 p.u. The p.u. value for the base values 13.8 kV and 50 MVA will be:	CO1	1
	a) 0.306 p.u. b) 0.33 p.u. c) 0.318 p.u. d) 0.328 p.u.		
	The positive, negative and zero sequence impedances of a solidly grounded system under steady state condition always follow the relations:	CO1	1
	(a) $Z_1 > Z_2 > Z_0$ (b) $Z_1 < Z_2 < Z_0$ (c) $Z_0 < Z_1 < Z_2$ (d) None of the above.		
4	The per unit impedance of a circuit element is 0.15. If the base kV and base MVA are halved, then the new value of the per unit impedance of the circuit element will be	CO2	1
	(a) 0.075 (b) 0.15 (c) 0.30 (d) 0.600.		
	A single line to ground fault occurs on an unloaded generator in phase a. If $x_d = x_2 = 0.25$ p.u., $x_0 = 0.15$ p.u., reactance connected in the neutral, $x_n = 0.05$ p.u. and the initial prefault voltage is 1.0 p.u., then the magnitude of the fault current will be	CO2	1
	(a) 3.75 p.u. (b) 1.54 p.u. (c) 1.43 p.u. (d) 1.25 p.u.		
6	Possible fault that may occur on a transmission line are	CO2	1
	a) 3 ph Fault b) LLG Fault c) LG fault d) LL Fault		

7	A transformer rated for 500 KVA 11 KV/0.4 KV has an impedance of 10% and is connected to an infinite bus. The fault level of the transformer is			CO3	1
	a) 500 KVA	b) 5000 KVA	c) $500\sqrt{3}$ KVA	d) 50 KVA	
8	The positive sequence component of voltage at the point of fault is zero when it is a			CO3	1
	a) 3 ph Fault	b) LLG Fault	c) LG fault	d) LL Fault	
	Zero sequence currents can flow from a line into a transformer bank if the windings are in			CO4	1
	a) grounded star/delta	b) delta/star	c) star grounded star	d) delta/delta	
10	Tick out the correct statement: (a) The negative and zero sequence voltages are maximum at the fault point and decrease towards the neutral (b) The negative and zero sequence voltages are minimum at the fault point and increase towards the neutral (c) The negative sequence is maximum and zero sequence minimum at the fault point and decrease and increase respectively towards the neutral. (d) None of the above.			CO4	1
	The bus admittance matrix (Ybus) of a power system is not			CO5	1
	(a) symmetric	(b) a square matrix	(c) a full matrix	d) dominant diagonal elements	
12	The bus admittance matrix of a power system is given as $\begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} -j50 & +j10 & +j5 \\ +j10 & -j30 & +j10 \\ +j5 & +j10 & -j25 \end{bmatrix} \end{matrix}$ The impedance of line between bus 2 and 3 will be equal to (a) $+j0.1$ (b) $-j0.1$ (c) $+j0.2$ (d) $-j0.2$			CO5	1
Q.12	Solve the following				12
A)	Define the term per unit voltage, per unit impedance and per unit voltamperes? What are the advantages of per unit representation.			CO1	6
B)	Draw per unit reactance diagram for three phase system shown in fig. choose 20 MVA and 66 KV as base value:			CO1	6

Q.3	Solve the following		12
A)	Classify various types of buses in a power system for load flow studies. Justify the classification.	CO2	6
B)	Determine admittance matrix it is given that all the line are characterized by series impedance $0.1 + j0.7 \text{ ohm/km}$ and shunt admittance of $j0.35 \times 10^{-5} \text{ mho/km}$.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Compare the performance of Gauss-Seidal method and Newton-Raphson method for load flow solution	CO3	6
B)	Explain sudden short circuit at the armature terminal of three phase generator at no load condition	CO3	6
C)	Write short note on Contingency Analysis.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Define α – operator ? draw the phaser representation of all sequence component with respective original system. and determine original phasors in terms of sequence component.	CO4	6
B)	A set of unbalanced line current in three phase, four wire system is as follows: $I_a = -j6 \text{ A}$, $I_b = (-8 + j5) \text{ A}$, $I_c = 7 \text{ A}$	CO4	6

	Determine zero, positive, negative sequence component of the current.		
C)	Determine phase voltage V_a, V_b and V_c from three phase unbalanced voltage in to symmetrical component gave the following result: $V_{a0} = 30\angle -30^\circ, V_{a1} = 450\angle 0^\circ$ and $V_{a2} = 225\angle 40^\circ$	CO4	6
Q4	Solve Any Two of the following.		12
A)	Derive the expression for fault current in case of LL fault considering the sequence network under this type of fault with suitable diagram.	CO5	6
B)	The star point of a 3 kV, 3 MVA three-phase synchronous generator is solidly grounded. Its positive-, negative- and zero-sequence reactances are 2.4 ohm, 0.45 ohm and 0.30 ohm respectively. The generator, operating unloaded, sustains a resistive fault between the a phase and ground. This fault has a resistance of 1.2 ohm. Calculate the fault current and the voltage to ground of a phase.	CO5	6
C)	What are Power quality causes and explain it?	CO5	6
*** End ***			